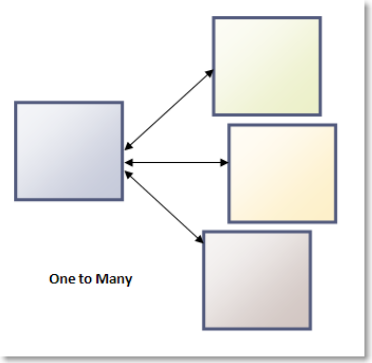
© Ahmad Esmaili, Robert F. Kelly

THE ONE-TO-MANY RELATIONSHIP

Reading: Chapter 4 (pages 89-95)



ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

Objectives

- Understand the one-to-many relationship between two entities
- Model a one-to-many relationship using an automated tool
- Define a database with a one-to-many relationship

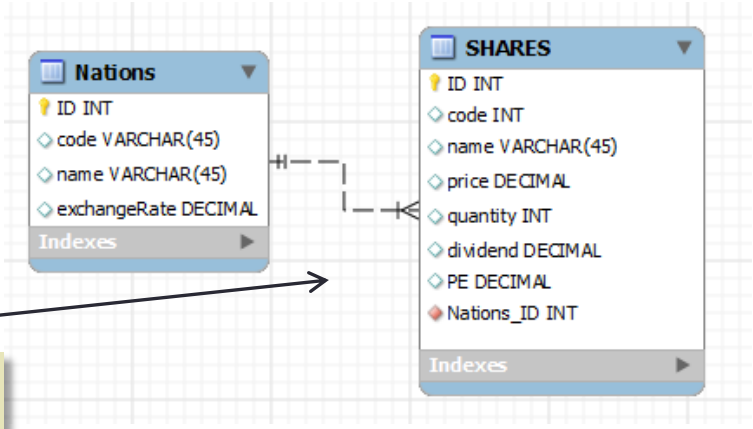
DB Design Diagrams

- Once we go beyond single table DBs, diagrams are helpful in design/specification
- Variations in level of diagram
  - Entity Relationship Diagram (ERD)
    - Higher level (entities, not tables)
    - Independent of DB implementation (relational, hierarchical, etc.)
    - Falling out of favor – textbooks favor it, but industry does not
  - Relational Data Model (RDM)
    - Assumes relational implementation
    - Deals with tables and relations
- Variation in notation
  - Dependent on automated tool used

The One-to-Many Relationship

- Entities are related to other entities
- One-to-many is usually shortened to 1:m
- Example
  - Shares of stock are listed on an exchange in a country
  - One country may have many stocks listed
  - A given stock is listed in only one country

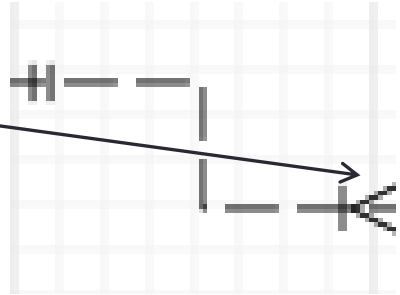
The connector between the two entities shows the relationship



Workbench notation

Relationship

- Identifies the multiplicity between two entities (e.g., one-to-one, one-to-many, many-to-many)
- Illustrated in the RDM
- We use the crows feet notation for relationships
  - Many side shows a crows foot
  - One side shows a simple connector
  - Vertical lines show minimum number



Helpful if ERDs resemble OO models

Text does not show minimum

ER Modeling

- Provide a visual representation of data
- Top-down approach to database design
- Identify
  - Entities (implemented as tables)
  - relationships between the entities
- Add details, such as
  - information about the entities and relationships
  - constraints on the entities, relationships, and attributes.

Our emphasis is primarily notation, with emphasis on concrete design (entity modeling is typically done through OO SW design)

Multiplicity Constraints

- Multiplicity constraints on relationships
  - Number of occurrences of one entity that may relate to a single occurrence of an associated entity

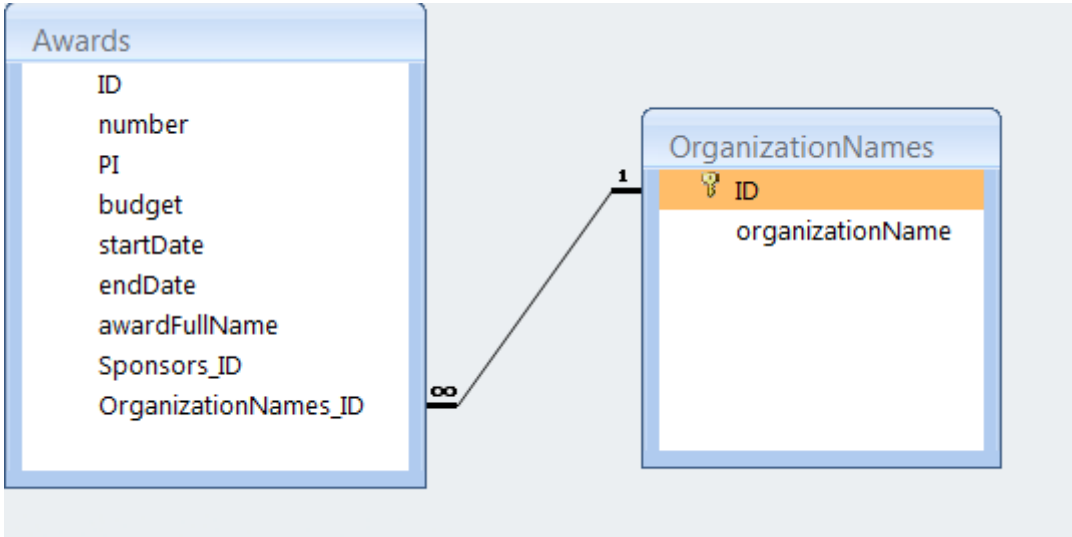
E.g., One nation lists many stocks

E-R Notation

- Chen
- IDEF1X
- Bachman
- Crow's Foot
- ISO
- UML

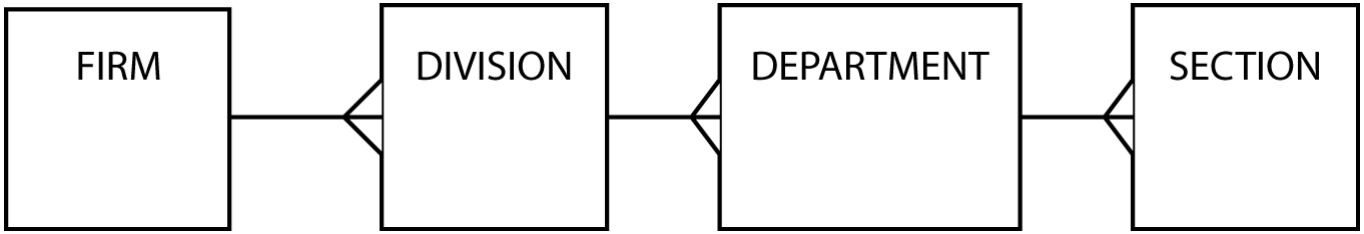
Some data modeling techniques (e.g., Chen) falling into disuse as OO modeling (e.g., UML) dominates

Access diagrams in Design View use a different notation



Hierarchical Relationships

- Form of 1-m relationship
- Occurs frequently
- Multiple 1:m relationships



Example – Shares DB

- Techniques to represent the Nations information
  1. Include Nation information in the Shares entity
  2. Provide Nation as a separate entity
- Including Nations information in the Shares entity might cause anomalies

Single Table Approach

- Add columns for nationName and exchangeRate

NationShares

ID	code	shareName	price	quantity	dividend	pe	nationName	exchangeRate
1	FC	Freedonia Copper	$27.50	10529	$1.84	16	UK	1
2	PT	Patagonian Tea	$55.25	12635	$2.50	10	UK	1
3	AR	Abyssinian Ruby	$31.82	22010	$1.32	13	UK	1
4	SLG	Sri Lankan Gold	$50.37	32868	$2.68	16	UK	1
5	ILZ	Indian Lead & Zinc	$37.75	6390	$3.00	12	UK	1
6	BE	Burmese Elephant	$0.07	154713	$0.01	3	UK	1
7	BS	Bolivian Sheep	$12.75	231678	$1.78	11	UK	1
8	NG	Nigerian Geese	$35.00	12323	$1.68	10	UK	1
9	CS	Canadian Sugar	$52.78	4716	$2.50	15	UK	1
10	ROF	Royal Ostrich Farms	$33.75	1234923	$3.00	6	UK	1
11	MG	Minnesota Gold	$53.87	816122	$1.00	25	USA	0.67
12	GP	Georgia Peach	$2.35	387333	$0.20	5	USA	0.67
13	NE	Narembeen Emu	$12.34	45619	$1.00	8	Australia	0.46
14	QD	Queensland Diamond	$6.73	89251	$0.50	7	Australia	0.46

Update Anomalies

- Tables that contain redundant information may potentially suffer from update anomalies
- Types of update anomalies include:
  - Insertion – how do you insert details of a new nation (e.g., exchange rate) that has no stock?
  - Deletion – when we delete the last stock, we lose the information about the nation
  - Modification – changes to a nation must be made for all records containing that nation

We will deal with this issue in more detail later in the semester

Mapping to a Relational Database

- Each entity becomes a table
- The entity name becomes the table name
- Each attribute becomes a column
- Add a column to the table at the many end of a 1:m relationship
- Put the identifier of the one end in the added column

Nations Table

- We create a new table using the definition of the Nations entity

NationShares | **Nations**

ID	code	nationName	exchangeRate
1	UK	United Kingdom	1.00
2	USA	United States	0.67
3	AUS	Australia	0.46
4	IND	India	0.02

India exchange rate is displayed with 2 digits of precision, but stored as .0228

Foreign Keys

- The 1:m relationship between 2 entities is implemented in a DB with a foreign key
- A foreign key is a column whose values are the primary keys of another table

Foreign Key

- nationID in the new Shares table is a foreign key that refers to the Nations table

Shares | Nations

ID	code	nationName	exchangeRate
1	UK	United Kingdom	1.00
2	USA	United States	0.67
3	AUS	Australia	0.46
4	IND	India	0.02

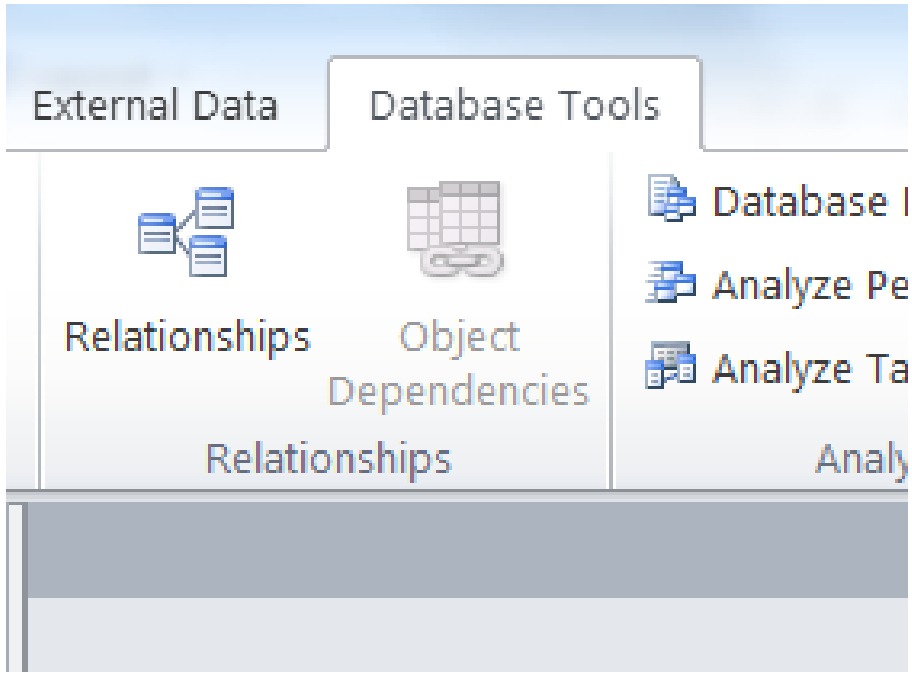
		shareName	price	quantity	dividend	pe	nationID
		Freedonia Copper	$27.50	10529	$1.84	16	1
		Patagonian Tea	$55.25	12635	$2.50	10	1
		Abyssinian Ruby	$31.82	22010	$1.32	13	1
		Sri Lankan Gold	$50.37	32868	$2.68	16	1
		Indian Lead & Zinc	$37.75	6390	$3.00	12	1
6	BE	Burmese Elephant	$0.07	154713	$0.01	3	1
7	BS	Bolivian Sheep	$12.75	231678	$1.78	11	1
8	NG	Nigerian Geese	$35.00	12323	$1.68	10	1
9	CS	Canadian Sugar	$52.78	4716	$2.50	15	1
10	ROF	Royal Ostrich Farms	$33.75	1234923	$3.00	6	1
12	MG	Minnesota Gold	$53.87	816122	$1.00	25	2
13	GP	Georgia Peach	$2.35	387333	$0.20	5	2
14	NE	Narembeen Emu	$12.34	45619	$1.00	8	3
15	QD	Queensland Diamond	$6.73	89251	$0.50	7	3
16	IR	Indooroopilly Ruby	$15.92	56147	$0.50	20	3
17	BD	Bombay Duck	$25.55	167382	$1.00	12	4

Referential Integrity Constraint

- For every value of a foreign key there is a primary key with that value
  - E.g., for every value of **`nationID`** in Shares there is a value of **`ID`** in **`Nations`**)
- A foreign key can never be null
- A primary key must exist before the foreign key can be defined
  - Must create Nations before Shares

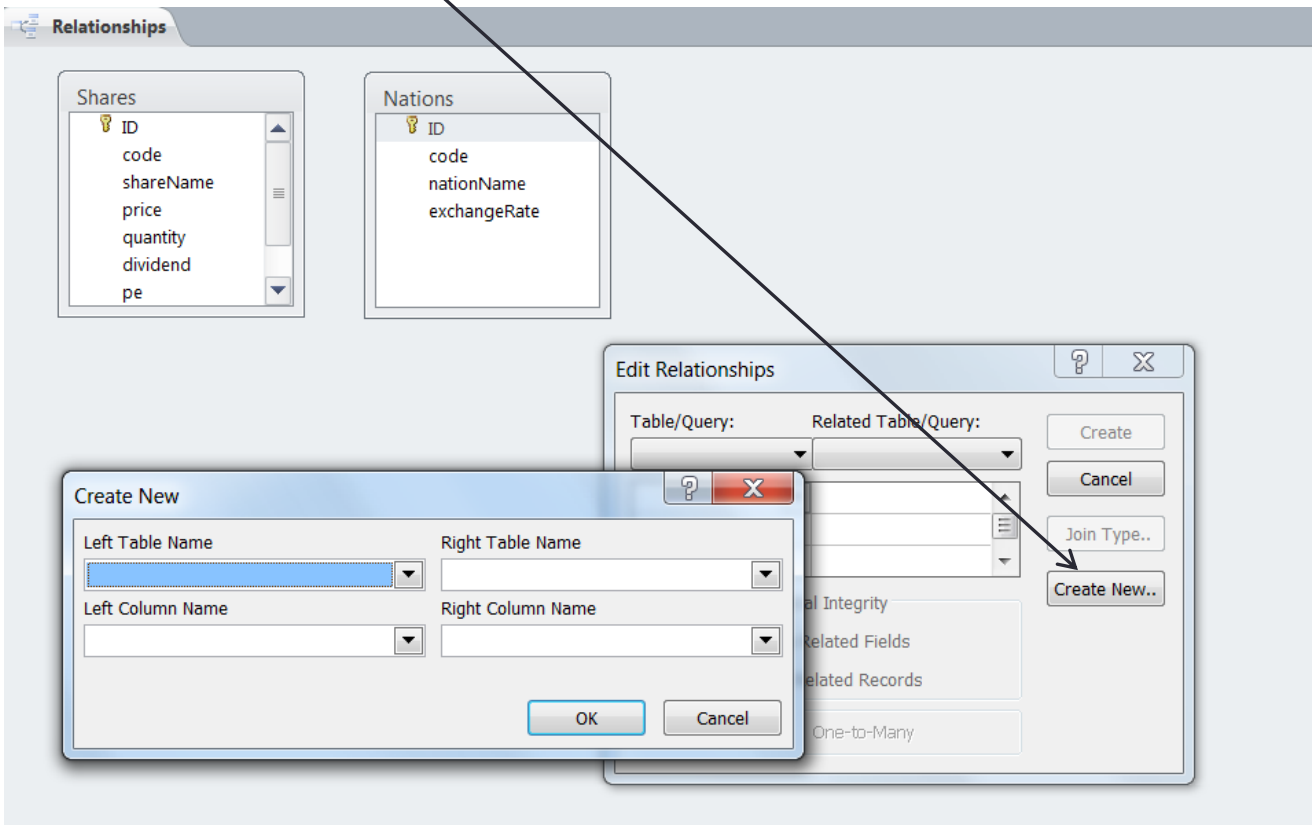
Relationships in Access

- To create a relationship in Access, first click on Relationships in Database Tools tab



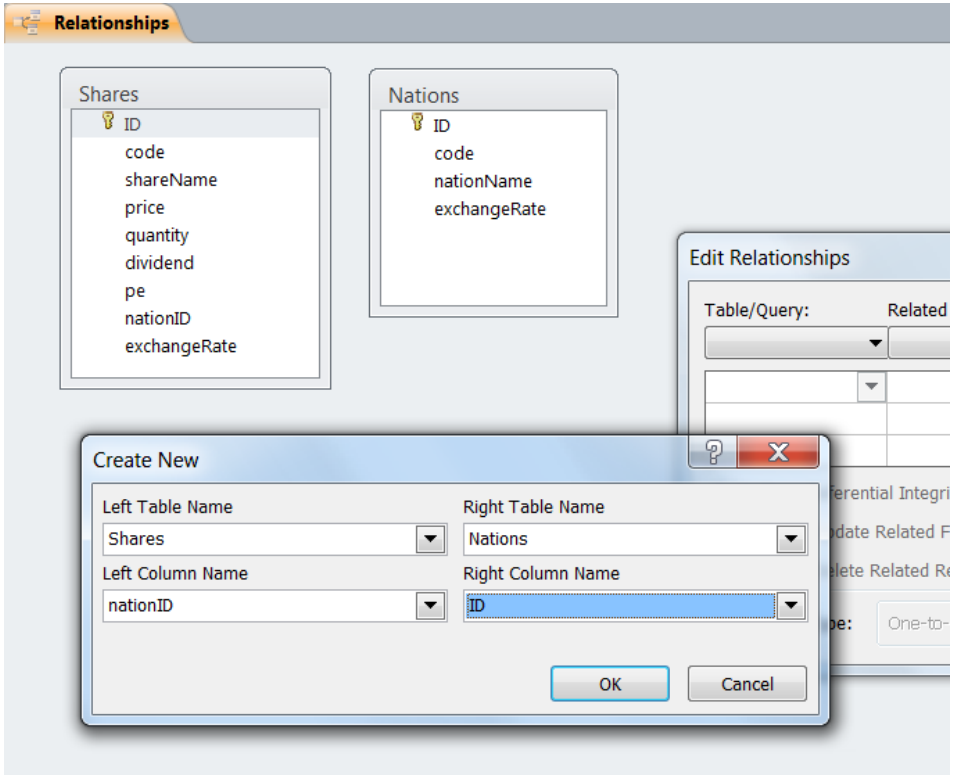
Relationships in Access

- Then click on Create New … to bring up the Create New dialog box



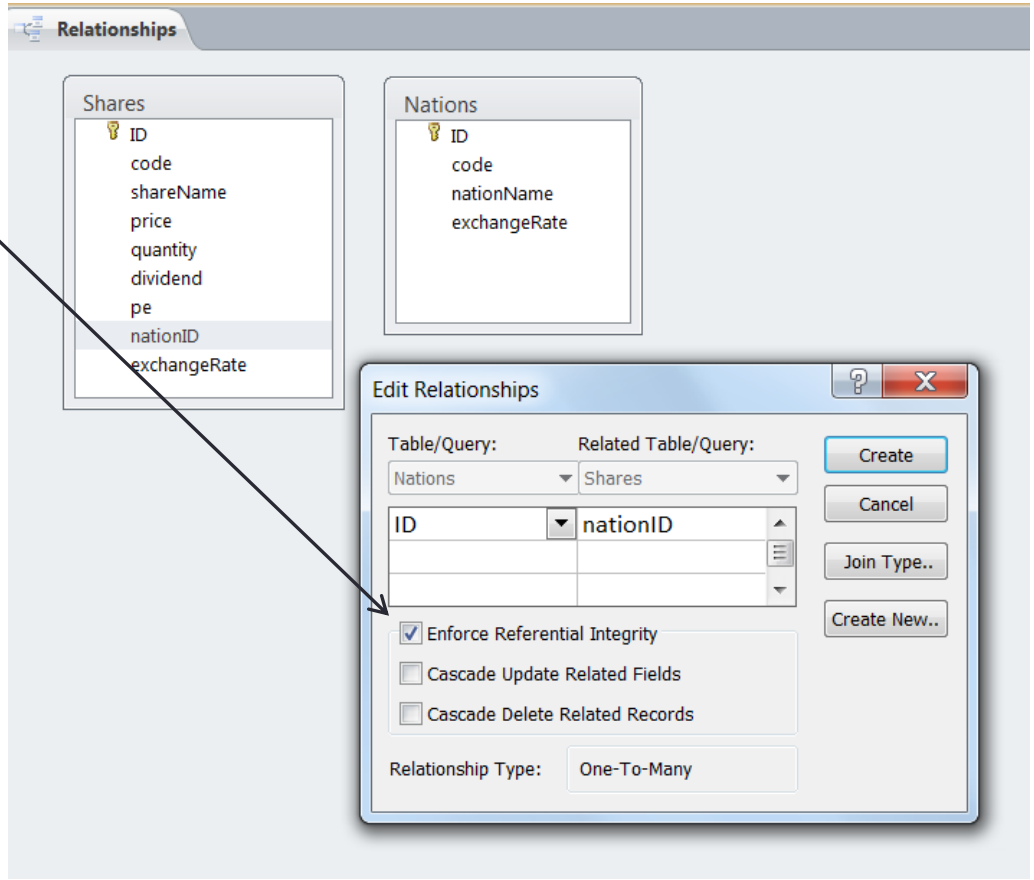
Relationships in Access

- For every value of a foreign key there is a primary key with that value



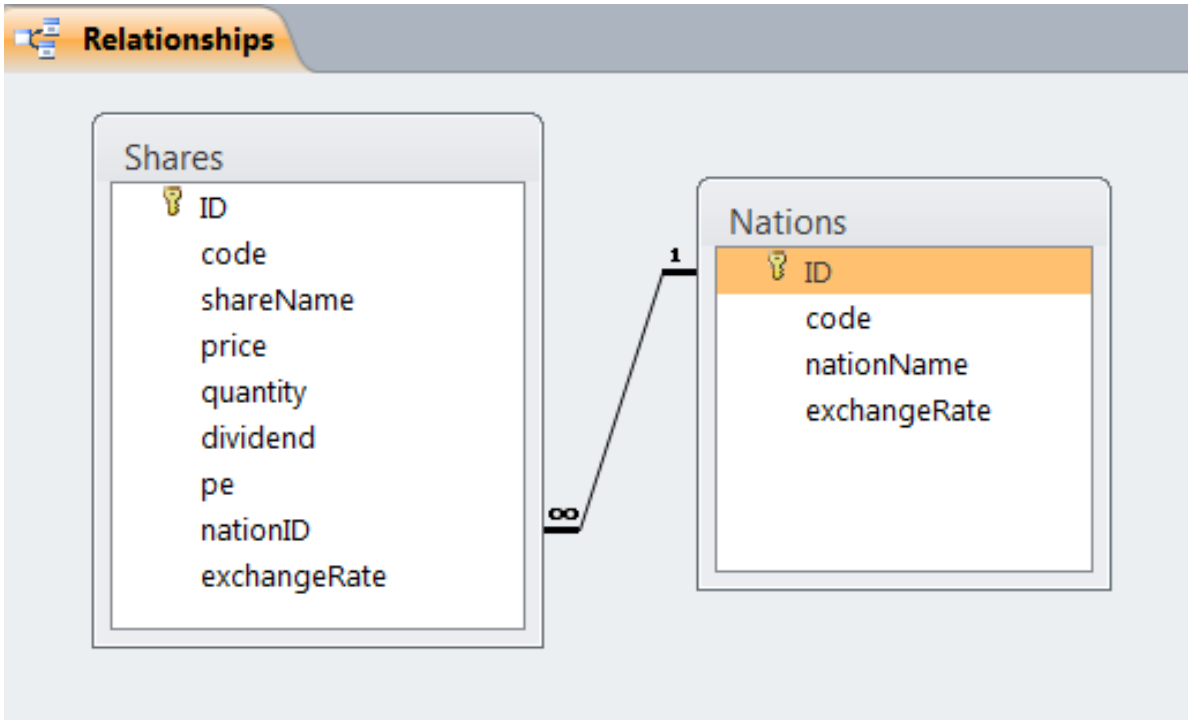
Relationships in Access

- Enforce referential integrity



Design View – Multiple Tables

- Access will show the relationships in Design View



Are We on Track?

- Create a new DB with two tables (Shares and Nations)
  - Create the Nations table with the following data
  - Add a foreign key to the Shares table using data in slide 11

NationShares | **Nations**

ID	code	nationName	exchangeRate
1	UK	United Kingdom	1.00
2	USA	United States	0.67
3	AUS	Australia	0.46
4	IND	India	0.02

Did You Meet the Objectives?

- Model a one-to-many relationship between two entities
- Model a one-to-many relationship using an automated tool
- Define a database with a one-to-many relationship